02. Dataset Understanding

Understand the source tables, join key and information used by the recommender.

Project: Movie Recommender System

Source archive supplied in this conversation: Movie-recommender-system-main.zip | Notebook: notebook.ipynb | App: app.py

Source data

Source	Important fields
tmdb_5000_movies.csv	id, title, overview, genres, keywords and other movie metadata
tmdb_5000_credits.csv	movie_id, title, cast, crew
Selected modelling fields	movie_id, title, overview, genres, keywords, cast, crew

Notebook observations

- The movies table has 4,803 rows and 20 columns before joining.
- The credits table includes movie_id, title, cast and crew.
- The notebook merges the two tables using title, not a unique numeric ID.
- The post-merge modelling table is not explicitly shape-checked in saved notebook output; later vector shape is (4,806, 5,000).
- The supplied ZIP contains only notebook.ipynb, app.py and README.md; the two CSV source datasets and generated pickle files are not included.

Why metadata fields matter

- Overview contributes plot/story terms.
- Genres contribute broad category signals such as Action, Drama or Comedy.
- Keywords contribute themes and topics.
- Cast contributes actor-name tokens; only the first three cast entries are retained.
- Crew is reduced to directors, which can encode stylistic similarity.

Data quality and join caveats

Merging on title can create duplicate or mismatched rows when titles are repeated or formatted differently. Prefer a stable unique key such as TMDB movie ID and validate one-to-one or many-to-one join cardinality. The observed vector shape of 4,806 rows, compared with 4,803 movie rows, is a signal to investigate the join and duplicate titles.

Missingness

The notebook calls dropna(inplace=True) on the selected columns before parsing metadata. This removes any movie with a missing selected field, but no missingness report or retained-row count is saved. An empty overview or empty cast/crew could also be valid data and should be distinguished from malformed JSON.

Data questions

- Are the source datasets the standard TMDB 5000 snapshots, and what are their licenses/terms?
- Should original language be used to filter or weight recommendations?
- How should remakes, franchises and duplicate titles be handled?
- Do we want similarity in plot, genre, cast, director, or a weighted blend?
- What should happen when a selected title is absent or has sparse metadata?